TRANSLATES OF AN OPEN BY A MORPHISM HAVE OPEN IMAGE IN THE ARROW SCHEME OF A GROUPOID OVER A FIELD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Base change to a field extension realizing points of the restricted groupoid (more-groupoids chapter, Lemma lemma-groupoid-on-field-explain-points, canonical tag 04LR; collected on this lane as L22505). cited at proof lines 1422-1424. Provides the extension k'/k and the point r' of R' over r lifting the given arrow, and enables the lift z'.
- Translates of opens by the composition map are open in the restricted groupoid over a field (more-groupoids chapter, Lemma lemma-groupoid-on-field-move-point, canonical tag 04LS; collected on this lane as L22506). cited at proof lines 1426-1428. Gives the open neighbourhood c(r', W') of r' contained in the translate image over R', the engine of the lifting argument.
- The restriction map R' to R is open for groupoids over a field (more-groupoids chapter, Lemma lemma-restrict-groupoid-on-field, canonical tag 04LQ). cited at proof lines 1433-1435. Pushes the open neighbourhood of the lifted point down to an open neighbourhood of the image of xi inside the original image.

Source reference. Original lemma 04LR: sources/stacks-chow/tags.txt, source lines 6017–6017. Cited at proof lines 1422-1424: provides the field extension k'/k and the point r' of R' lifting r, enabling the lift z'.

Source reference. Original lemma 04LS: sources/stacks-chow/tags.txt, source lines 6018–6018. Cited at proof lines 1426-1428: c(r', W') is an open of R' contained in the translate image over R'.

Source reference. Original lemma 04LQ: sources/stacks-chow/tags.txt, source lines 6016–6016. Cited at proof lines 1433-1435: R' to R is open, pushing the lifted open neighbourhood down to the original image.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. If U is the spectrum of a field, $W \subset R$ is open, and $Z \to R$ is a morphism of schemes, then the image of the composition $Z \times_{s,U,t} W \to R \times_{s,U,t} R \to R$ is open.*

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Proof. Write $U = \mathrm{Spec}(k)$. Consider a field extension k'/k. Denote $U' = \mathrm{Spec}(k')$. Let R' be the restriction of R via $U' \to U$. Set $Z' = Z \times_R R'$ and $W' = R' \times_R W$. Consider a point $\xi = (z, w)$ of $Z \times_{s,U,t} W$. Let $r \in R$ be the image of z under $Z \to R$. Pick $k' \supset k$ and $r' \in R'$ as in Lemma 04LR. We can choose $z' \in Z'$ mapping to z and r'. Then we can find $\xi' \in Z' \times_{s',U',t'} W'$ mapping to z' and ξ. The open $c(r', W')$ (Lemma 04LS) is contained in the image of $Z' \times_{s',U',t'} W' \to R'$. Observe that $Z' \times_{s',U',t'} W' = (Z \times_{s,U,t} W) \times_{R \times_{s,U,t} R} (R' \times_{s',U',t'} R')$. Hence the image of $Z' \times_{s',U',t'} W' \to R' \to R$ is contained in the image of $Z \times_{s,U,t} W \to R$. As $R' \to R$ is open (Lemma 04LQ) we conclude the image contains an open neighbourhood of the image of ξ as desired. $\square$